



Modelling Financial Markets

Chapter 11: Continuous-Time Models



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BSS: Brownian SemiStationary (process)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CBOE: Chicago Board Options Exchange
- ▣ CI: Confidence Interval
- ▣ CIR: Cox–Ingersoll–Ross (model)
- ▣ CKLS: Chan–Karolyi–Longstaff–Sanders (model)
- ▣ CLT: Central Limit Theorem
- ▣ DGS10: 10-Year Treasury Constant Maturity Rate (FRED series code)
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GBM: Geometric Brownian Motion
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ JB: Jarque–Bera (test)
- ▣ LLM: Large Language Model
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error

Acronyms used in this material (2/2)

- ▣ **NLLS: Non-Linear Least Squares**
- ▣ **OLS: Ordinary Least Squares**
- ▣ **OU: Ornstein–Uhlenbeck (process)**
- ▣ **PDE: Partial Differential Equation**
- ▣ **QLIKE: Quasi-LIKelihood (loss function)**
- ▣ **QQ: Quantile–Quantile (plot)**
- ▣ **RFSV: Rough Fractional Stochastic Volatility (model)**
- ▣ **SDE: Stochastic Differential Equation**
- ▣ **SVJ: Stochastic Volatility with Jumps (model)**
- ▣ **US: United States**
- ▣ **VIX: CBOE Volatility Index**
- ▣ **XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)**

Outline

Motivation

From Random Walk to Brownian Motion

Quadratic Variation and Itô Calculus

Stochastic Differential Equations and Simulation

Geometric Brownian Motion

Mean Reversion: the Ornstein–Uhlenbeck Process

Estimating Diffusions: What Is Identified from What

Change of Measure: Girsanov and Feynman–Kac

Jumps: the Merton Model

Stochastic Volatility: the Heston Model

Models against the Data

Case Study: Rough and Persistent Volatility

AI for Scientific Discovery

Summary

References

Why Continuous Time?

The question of this chapter

- Prices are recorded daily or every few milliseconds, but trading never really stops
- Can we describe prices with a model that works at **every** time scale at once?
- Option pricing (Chapter 12), interest-rate models and risk management are written in this language

What we do in this chapter

- Build Brownian motion and Itô calculus through simulation
- Simulate stochastic differential equations and measure the error of the schemes
- Estimate four models on data: geometric Brownian motion, Ornstein–Uhlenbeck, Merton jumps, Heston volatility
- Confront every model with the stylised facts of Chapter 1

From Brown to Bachelier, Einstein and Wiener



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Albert Einstein in 1905

Photo: Lucien Chavan / ETH Zürich (1905); public domain; Wikimedia Commons

- 1827: the botanist Robert Brown sees pollen particles move in an irregular zig-zag under the microscope
- Bachelier (1900): French government bond (rente) prices and their options follow a random walk in continuous time
 - ▶ The spread of the price grows with $\sqrt{t}$, five years before physics
- Einstein (1905): molecular collisions explain Brown's motion; the mean squared displacement is linear in t
- Wiener (1923): Brownian motion as a probability measure on continuous paths, the **Wiener process**

Itô, Samuelson, Black, Scholes and Merton



Kiyosi Itô (1915–2008)

Photo: Konrad Jacobs (1970); CC
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Paul A. Samuelson (1915–2009)

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- Itô (1944): the stochastic integral and its chain rule, the main working tool of this chapter
- Osborne (1959): log prices, not prices, behave like Brownian motion; Samuelson (1965): geometric Brownian motion
- Black & Scholes (1973) and Merton (1973): an option can be replicated in continuous time
- Merton (1976) adds jumps, Heston (1993) random volatility; textbook treatment: Franke, Härdle & Hafner (2019), Part I

Data Used in This Chapter

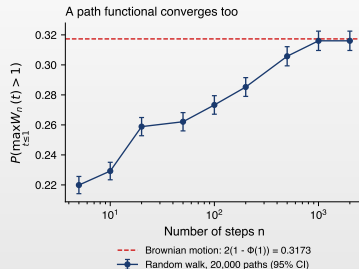
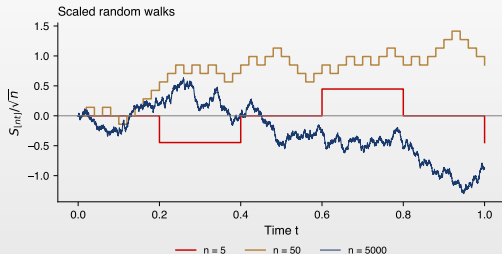
Series	Type	Period	Days
S&P 500	US equity index	1990–2026	9 245
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
VIX	CBOE implied volatility index of the S&P 500	1990–2026	daily
DTB3	3-month US Treasury bill rate (FRED)	1954–2026	daily

- Daily closing values up to 18 September 2026; daily log returns $r_t = \ln P_t - \ln P_{t-1}$
- Each series on its own calendar; equity indices on weekdays, stale quotes removed (unchanged close and no intraday range); genuine zero returns kept
- Time in years, daily step $\Delta t = 1/252$ (Bitcoin: 1/365); drifts and volatilities annualised
- W_t : standard Brownian motion; S_t : price; v_t : instantaneous variance; $\mathcal{F}_t$: the information (filtration) up to t

From Random Walk to Brownian Motion: Donsker's Theorem

- Scaled random walk $W_n(t) = S_{\lfloor nt \rfloor} / \sqrt{n}$, $S_k = \xi_1 + \dots + \xi_k$, steps i.i.d. with mean 0 and variance 1
 - ▶ Time step $1/n$, space step $1/\sqrt{n}$: the only scaling with a non-trivial limit; by the central limit theorem (CLT), $W_n(t) \rightarrow N(0, t)$ for each t
- Donsker: $W_n \Rightarrow W$ as random functions, i.e. for the **whole path** Billingsley (1999)
 - ▶ “Invariance”: the limit does not depend on the distribution of the steps
- Continuous mapping: every continuous functional of the path converges too
 - ▶ Example: the maximum; by the reflection principle $P(\max_{t \leq 1} W_t > a) = 2P(W_1 > a)$
- For finance: many small independent trades add up to Brownian-like price paths

Random Walks Converge to Brownian Motion

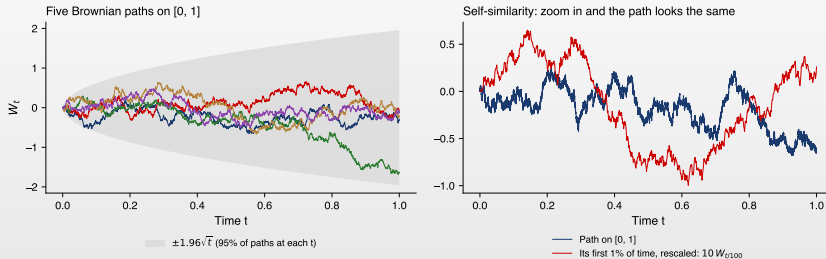


- Left: one scaled random walk for $n = 5, 50, 5000$ steps; right: probability that the path exceeds 1 before $t = 1$
- $P(\max W_n > 1)$ rises from 0.220 ($n = 5$) to 0.316 ($n = 2000$), against the Brownian $2(1 - \Phi(1)) = 0.3173$ (Φ : standard Normal distribution function)
- A coarse grid misses the maximum between observations, and convergence is slow, of order $1/\sqrt{n}$: daily closes understate the intraday range

Brownian Motion: Definition and Properties

- **Standard Brownian motion:** $W_0 = 0$; independent increments; $W_t - W_s \sim N(0, t - s)$ for $s < t$; continuous paths
 - ▶ Existence: [Wiener \(1923\)](#); simulation on a grid: $W_{t+\Delta t} = W_t + \sqrt{\Delta t} Z$, $Z \sim N(0, 1)$
- $\text{Cov}(W_s, W_t) = \min(s, t)$; a Gaussian process and a **martingale**: $\mathbb{E}[W_t | \mathcal{F}_s] = W_s$ (the efficient-market random walk of Chapter 2)
- **Self-similarity:** $\{c^{-1/2} W_{ct}\}$ is again a standard Brownian motion for any $c > 0$
- Paths are continuous but **nowhere differentiable**
 - ▶ $(W_{t+h} - W_t)/h$ has standard deviation $1/\sqrt{h} \rightarrow \infty$: ordinary calculus fails

Brownian Paths

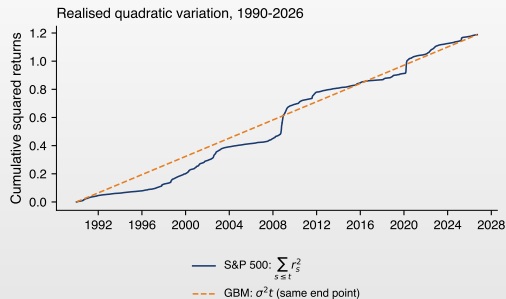
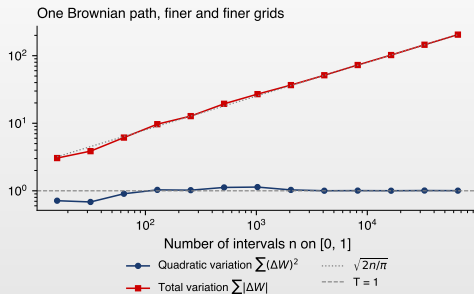


- Left: five paths with the band $\pm 1.96\sqrt{t}$; right: a path and its first 1% of time, stretched 100 times in time and 10 times in space
- The band widens like $\sqrt{t}$ (the square-root-of-time rule of Chapter 7); a zoomed piece looks like the whole path
- Real markets break self-similarity: fat tails and clustering depend on the horizon (Chapter 1)

Total and Quadratic Variation

- Grid $0 = t_0 < t_1 < \dots < t_n = T$ with step $\Delta t = T/n$ and increments $\Delta W_k = W_{t_k} - W_{t_{k-1}}$
- **Total variation** $\sum_k |\Delta W_k|$: the length of the path
 - ▶ Its mean is $\sqrt{2nT/\pi} \rightarrow \infty$: a Brownian path has infinite length
- **Quadratic variation** $[W]_T = \lim \sum_k (\Delta W_k)^2 = T$
 - ▶ A random sum with a **deterministic** limit
 - ▶ $\text{Var}(Q_n) = 2T^2/n$: one year of daily data gives about 9% relative error ($\sqrt{2/252}$), the idea behind realised variance (Chapter 9)
- Shorthand used in Itô calculus: $(dW_t)^2 = dt$

Quadratic Variation: Simulation and Data



□ Left: one Brownian path on grids with 16 to 65,536 intervals; right: cumulative sum of squared daily S&P 500 log returns

 MFM_ch11_quadratic_variation

Reading the Quadratic Variation Chart

- ▣ Simulation: the quadratic variation settles at 1.006 ($T = 1$); the total variation keeps growing, 205 against $\sqrt{2n/\pi} = 204$
- ▣ S&P 500: total $\sum r_t^2 = 1.188$ over 36.7 years, an average volatility of 18.0%
 - ▶ Under geometric Brownian motion the curve would be a straight line
- ▣ It moves in steps: September 2008 – March 2009 is 1.6% of the days but 15.2% of all quadratic variation
 - ▶ February – April 2020: 0.5% of the days, 7.8% of the quadratic variation
- ▣ Volatility is itself random: the motivation for the Heston model at the end of the chapter

The Itô Integral

- Goal: give a meaning to $\int_0^T H_t dW_t$, the gain of holding H_t units of an asset whose price moves like W
 - ▶ Riemann–Stieltjes integration fails because the path has infinite total variation
- **Itô integral**: the limit of sums with the integrand at the **left** end point
 - ▶ $\int_0^T H_t dW_t = \lim_{n \rightarrow \infty} \sum_{k=1}^n H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}})$ (limit in mean square)
- H must be **adapted** (progressively measurable) and square-integrable, $\mathbb{E} \left[\int_0^T H_t^2 dt \right] < \infty$
 - ▶ Financial reading: H_t uses only information up to t ; you choose the position before you see the next price move
 - ▶ Left-point sums converge for regular integrands (e.g. continuous and adapted); a general H is reached by approximation with simple processes

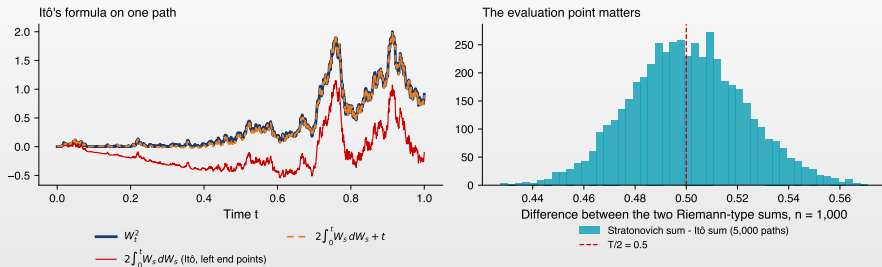
Properties of the Itô Integral

- For square-integrable adapted H (previous slide):
- Zero mean: $\mathbb{E} \left[\int_0^T H_t dW_t \right] = 0$: no strategy earns money from a martingale on average
- **Itô isometry**: $\mathbb{E} \left[\left(\int_0^T H_t dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T H_t^2 dt \right]$
- The process $t \mapsto \int_0^t H_s dW_s$ is a martingale
 - ▶ If only $\int_0^T H_t^2 dt < \infty$ almost surely, it is merely a local martingale, and its mean need not be zero
- The evaluation point matters: the mid-point rule gives the **Stratonovich** integral, a different number
 - ▶ Finance uses Itô, because a mid-point position would look into the future

Worked Example: $\int_0^T W_t \, dW_t$

- ▣ Identity for each term: $W_{t_{k-1}} \Delta W_k = \frac{1}{2} (W_{t_k}^2 - W_{t_{k-1}}^2) - \frac{1}{2} (\Delta W_k)^2$
- ▣ Sum over k : the first part telescopes to $\frac{1}{2} W_T^2$, the second tends to $\frac{1}{2} T$
 - ▶ $\int_0^T W_t \, dW_t = \frac{1}{2} W_T^2 - \frac{1}{2} T$
- ▣ Ordinary calculus would give $\frac{1}{2} W_T^2$; the extra $-\frac{1}{2} T$ is the quadratic variation
- ▣ Check: the mean is $\frac{1}{2}(T - T) = 0$, as it must be for an Itô integral

Itô's Formula on Simulated Paths



- Left: W_t^2 , the Itô integral $2 \int_0^t W \, dW$ and their difference t on one path; right: Stratonovich minus Itô sums over 5,000 paths
- $2 \int_0^t W \, dW + t$ lies exactly on W_t^2 : the formula holds path by path; Stratonovich minus Itô: mean 0.500, s.d. 0.022 (theory $T/2 = 0.5$)
- Itô sums: mean 0.007, variance 0.520; theory: 0 and, by the isometry, $\int_0^1 t \, dt = 1/2$

Itô Processes and Itô's Lemma

- **Itô process:** $dX_t = a_t dt + b_t dW_t$ (drift a_t , diffusion coefficient b_t , both adapted)
 - ▶ Shorthand for $X_t = X_0 + \int_0^t a_s ds + \int_0^t b_s dW_s$
- **Itô's lemma:** for $f(t, x)$ twice differentiable [Itô \(1944\)](#)
 - ▶ $df(t, X_t) = (f_t + a_t f_x + \frac{1}{2} b_t^2 f_{xx}) dt + b_t f_x dW_t$
- Rule of thumb: Taylor expansion to second order with $dt \cdot dt = 0$, $dt \cdot dW = 0$, $dW \cdot dW = dt$
 - ▶ The second-order term $\frac{1}{2} b_t^2 f_{xx} dt$ does not vanish: this is the whole novelty

Worked Example: the Logarithm of a Price

- Price dynamics $dS_t = \mu S_t dt + \sigma S_t dW_t$; take $f(x) = \ln x$, so $f_x = 1/x$, $f_{xx} = -1/x^2$
- Itô's lemma with $a_t = \mu S_t$, $b_t = \sigma S_t$
 - ▶ $d \ln S_t = \left(\mu S_t \cdot \frac{1}{S_t} - \frac{1}{2} \sigma^2 S_t^2 \cdot \frac{1}{S_t^2} \right) dt + \sigma S_t \cdot \frac{1}{S_t} dW_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t$
- The log price grows at $\mu - \sigma^2/2$, not at μ : volatility lowers the growth rate of the median investor's wealth
- With $\sigma = 18\%$: $\sigma^2/2 = 1.6$ percentage points per year; with $\sigma = 67\%$ (Bitcoin): 22 points

Quadratic Covariation and the Multivariate Itô Formula

- **Quadratic covariation** $[X, Y]_t = \lim \sum_k \Delta X_k \Delta Y_k$; if $\text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$, then $d[W^{(1)}, W^{(2)}]_t = \rho dt$
 - ▶ Product rule: $d(X_t Y_t) = X_t dY_t + Y_t dX_t + d[X, Y]_t$
- **Multivariate Itô formula** for $X_t \in \mathbb{R}^d$ and $f(t, x)$ twice differentiable
 - ▶ $df(t, X_t) = f_t dt + \sum_i f_{x_i} dX_t^i + \frac{1}{2} \sum_{i,j} f_{x_i x_j} d[X^i, X^j]_t$
- Worked example (Heston, Section 10): $dS = \mu S dt + \sqrt{v} S dW^{(1)}$, $dv = \kappa(\theta - v) dt + \xi \sqrt{v} dW^{(2)}$
 - ▶ $d(S_t v_t) = S_t dv_t + v_t dS_t + \rho \xi v_t S_t dt$
 - ▶ ρ enters the drift of the product: with $\rho < 0$, price and variance tend to move in opposite directions

Recap: Itô Calculus

- Brownian paths have infinite length but finite quadratic variation $[W]_t = t$
- The Itô integral uses left end points: for square-integrable integrands it is a martingale with zero mean
- Itô's lemma = Taylor to second order with $(dW)^2 = dt$; with several Brownian motions add $d[W^{(1)}, W^{(2)}] = \rho dt$
- Real prices have a quadratic variation that grows in bursts, not linearly

Stochastic Differential Equations

- A **stochastic differential equation** (SDE): $dX_t = a(t, X_t) dt + b(t, X_t) dW_t$, X_0 given
 - ▶ Drift a and diffusion b now depend on the current state
- Existence and uniqueness of a solution: a and b Lipschitz in x with at most linear growth [Kloeden & Platen \(1992\)](#)
- Few SDEs have closed-form solutions; the four models of this chapter are among them
 - ▶ Geometric Brownian motion, Ornstein–Uhlenbeck and Merton: exact; Heston: only the characteristic function
- Otherwise we simulate on a grid; the question is how large the discretisation error is

Euler–Maruyama and Milstein Schemes

- **Euler–Maruyama:** freeze a and b over each step [Higham \(2001\)](#)

- ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k, \Delta W_k = \sqrt{\Delta t} Z_k$

- **Milstein:** add the next Itô–Taylor term [Milstein \(1975\)](#)

- ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k + \frac{1}{2} b(X_k) b'(X_k) [(\Delta W_k)^2 - \Delta t]$

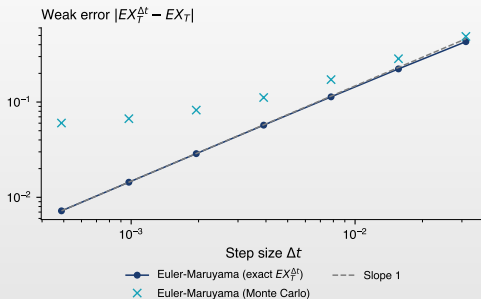
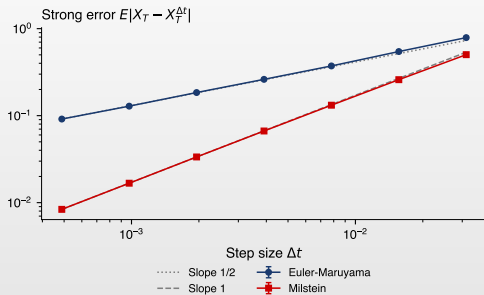
- The correction uses $(\Delta W_k)^2 - \Delta t$, which has mean zero: it only matters path by path

- Test equation [Higham \(2001\)](#): $dX = \lambda X dt + \mu X dW$, $\lambda = 2$, $\mu = 1$, $X_0 = 1$, $T = 1$, with the exact solution as benchmark

Strong and Weak Convergence

- **Strong error:** $\mathbb{E}|X_T - X_T^{\Delta t}| \leq C \Delta t^\gamma$, how close each simulated path is to the true one
 - ▶ Euler–Maruyama $\gamma = 1/2$; Milstein $\gamma = 1$
 - ▶ Needed for path-dependent quantities: hedging errors, barrier crossings
- **Weak error:** $|\mathbb{E}[g(X_T^{\Delta t})] - \mathbb{E}[g(X_T)]| \leq C \Delta t^\beta$, how close the distribution is
 - ▶ Both schemes: $\beta = 1$; enough for prices of European options, VaR, expected values
- On a log-log plot, the order is the slope of error against step size

Convergence of the Schemes



□ 20,000 Brownian paths on a fine grid of 2^{11} steps, coarsened to steps from 2^{-11} to 2^{-5} ; bars are 95% CIs

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Reading the Convergence Chart

- ▣ Estimated strong orders (slope of the log-log fit; 95% CI by resampling whole paths, all step sizes together): Euler–Maruyama 0.518 [0.512, 0.525], Milstein 0.986 [0.984, 0.987]
 - ▶ Close to the theoretical 1/2 and 1; the narrow intervals measure Monte Carlo noise only, and the small gap is finite-step curvature, not a different order
 - ▶ Halving Δt cuts the Milstein error in half, the Euler error only by 29%
- ▣ Weak order of Euler–Maruyama from the exact mean $(1 + \lambda \Delta t)^{T/\Delta t}$: 0.985, a deterministic slope with no sampling error
 - ▶ The Monte Carlo estimates flatten at small steps: their standard error, about 0.068, exceeds the bias
- ▣ In practice the sampling error of Monte Carlo, not the step size, is often the binding constraint [Glasserman \(2003\)](#)

Recap: Simulation

- An SDE specifies the local drift and diffusion of a process
- Euler–Maruyama is simple and strong order $1/2$; Milstein adds one term and reaches order 1
- For expected values the weak order 1 is what matters, together with the Monte Carlo sample size
- When the exact transition is known (next sections), simulate exactly: no discretisation error

Geometric Brownian Motion (GBM)

- SDE: $dS_t = \mu S_t dt + \sigma S_t dW_t$, i.e. the **percentage** change has constant drift and volatility
 - ▶ μ : expected rate of return; σ : volatility, both per year
- Solution (from the log example): $S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right)$
 - ▶ Prices stay positive; exact simulation needs only Normal draws
- The price model behind [Black & Scholes \(1973\)](#); proposed by [Osborne \(1959\)](#) and [Samuelson \(1965\)](#)
- Textbook treatment: [Franke, Härdle & Hafner \(2019\)](#), chapters on the Wiener process and stochastic integration

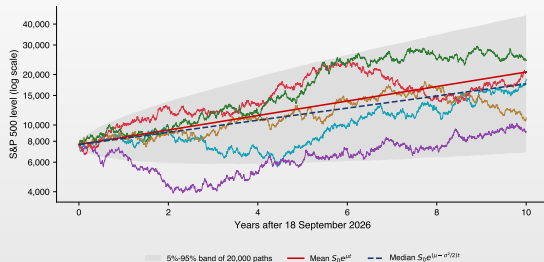
Log Returns under GBM

- Log returns over Δt : $r_t = \ln(S_t/S_{t-\Delta t}) \sim N((\mu - \frac{1}{2}\sigma^2)\Delta t, \sigma^2\Delta t)$
 - ▶ Independent and identically distributed (i.i.d.) at every frequency
- Testable implications
 - ▶ Normal log returns: skewness 0, excess kurtosis 0
 - ▶ No autocorrelation in r_t , $|r_t|$ or r_t^2
 - ▶ Variance proportional to the horizon
- S_T/S_0 is **lognormal**: $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$ with the log drift $m = \mu - \sigma^2/2$

Mean, Median and Mode of a Lognormal Price

- Mean $\mathbb{E}[S_T] = S_0 e^{\mu T}$; median $S_0 e^{mT}$; mode $S_0 e^{(m-\sigma^2)T}$: mode < median < mean
- Worked example, S&P 500 parameters: $\hat{\mu} = 10.0\%$, $\hat{\sigma} = 18.0\%$, $T = 10$ years
 - ▶ Mean gross return 2.71, median 2.30, mode 1.66
 - ▶ $P(S_T < S_0) = \Phi(-m\sqrt{T}/\sigma) = 7.1\%$
 - ▶ $P(S_T < \mathbb{E}[S_T]) = \Phi(\sigma\sqrt{T}/2) = 61.2\%$: most investors end **below** the average
- The average is pulled up by a few very large outcomes; the gap grows with $\sigma^2 T$ (Bitcoin: $\sigma \approx 67\%$)
- For long horizons, report the median and quantiles, not only the mean

Ten Years of Simulated S&P 500 Paths



- GBM with $\hat{\mu}$, $\hat{\sigma}$ from 1990–2026, starting at the close of 18 September 2026 ($S_0 = 7\,650$); 20,000 exact paths, five shown
- After ten years: mean 20 696, median 17 604, 5%–95% range 6 878 – 45 017; $P(S_T < S_0) = 7.1\%$
- Too narrow in crises and too smooth: GBM has no fat tails and no volatility clustering (tested below)

Estimating GBM by Maximum Likelihood

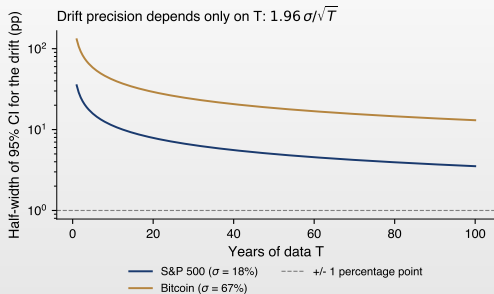
- Maximum likelihood estimation (MLE) from n log returns at step Δt , over $T = n\Delta t$ years
 - ▶ $\hat{\sigma}^2 = \frac{1}{n\Delta t} \sum_t (r_t - \bar{r})^2$, $\hat{m} = \bar{r}/\Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2$
- Standard errors under GBM (i.i.d. Normal returns): $\text{s.e.}(\hat{\sigma}) = \sigma/\sqrt{2n}$ and $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$
 - ▶ Volatility precision improves with more observations (sample more often)
 - ▶ Drift precision depends only on the **calendar length** T : daily or minute data do not help
- Reason: $\sum_t r_t = \ln(S_T/S_0)$ uses only the first and the last price

Worked Example: GBM for Three Markets

	Years	$\hat{\sigma}$ (%)	s.e.($\hat{\sigma}$)	$\hat{m}$ (%)	s.e.($\hat{m}$)
S&P 500	36.7	18.0	0.13	8.3	3.0
BET	26.5	22.2	0.19	16.1	4.3
Bitcoin	12.0	66.6	0.71	43.1	19.2

- Under GBM, volatility is pinned down to about 0.1–0.7 percentage points; the log drift only to 3–19 points
- These s.e. are conditional on GBM: with excess kurtosis k , even independent returns inflate s.e.($\hat{\sigma}$) by $\sqrt{(k+2)/2} = 2.5$ (S&P 500); block-bootstrap s.e. in Seminar B2
- S&P 500: $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2 = 8.3\% + 1.62\% = 10.0\%$ per year
- A 95% CI of ± 1 point for the S&P 500 drift would need $(1.96 \hat{\sigma}/0.01)^2 = 1243$ years of data

How Precisely Can We Estimate the Drift?



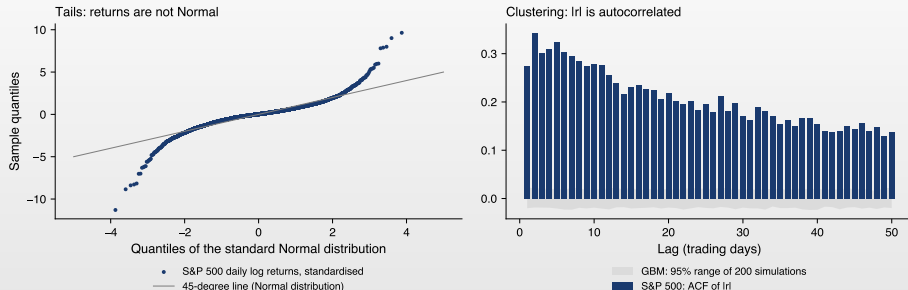
□ Left: half-width of the 95% CI for the drift against years of data; right: S&P 500 estimates on rolling 10-year windows

 MFM_ch11_drift_precision

Reading the Drift Chart

- Ten-year log drift of the S&P 500 ranges from -6.0% (window ending 2009-03) to 15.8% (2000-09)
 - ▶ Every window's 95% CI is about ± 11 points wide: the variation is mostly noise
- Ten-year volatility stays between 14.1% and 22.3% : estimated precisely, but not constant
- Bitcoin would need 17 028 years for ± 1 point: its expected return is essentially unknowable from its price history
- Option pricing in Chapter 12 avoids μ : under the risk-neutral measure only σ enters (Girsanov, Section 8)

Does GBM Fit the S&P 500?



- Left: quantile–quantile (QQ) plot of standardised daily log returns against the Normal distribution; right: autocorrelation function (ACF) of $|r_t|$ against 200 GBM samples of the same length

Reading the GBM Test

- Tails: excess kurtosis 10.9 instead of 0; the worst day is -11.3 standard deviations
 - ▶ 62 days beyond ± 4 standard deviations, against 0.59 expected under GBM
 - ▶ Jarque–Bera (JB) statistic 45 805: Normality is rejected at any level
- Clustering: ACF of $|r_t|$ is 0.27 at lag 1 and still 0.14 at lag 50
 - ▶ GBM samples never exceed 0.024 in absolute value
- GBM fails on both stylised facts of Chapter 1 [Cont \(2001\)](#); the rest of the chapter adds mean reversion, jumps and random volatility

Recap: Geometric Brownian Motion

- GBM: constant percentage drift and volatility; lognormal prices; i.i.d. Normal log returns
- The median grows at $\mu - \sigma^2/2$; the mean at μ ; most paths end below the mean
- Volatility is estimable from data; the drift is not, in any realistic sample
- The data reject GBM: fat tails and volatility clustering

The Ornstein–Uhlenbeck (OU) Process

- SDE: $dx_t = \kappa(\theta - x_t) dt + \sigma dW_t$, $\kappa > 0$ Uhlenbeck & Ornstein (1930)
 - ▶ θ : long-run mean; κ : speed of mean reversion; σ : volatility
 - ▶ Above θ the drift is negative, below θ positive: a spring pulling x back
- Used for interest rates, volatility, spreads, commodity prices, exchange-rate deviations
 - ▶ Vasicek (1977): the OU process as a model of the short-term interest rate

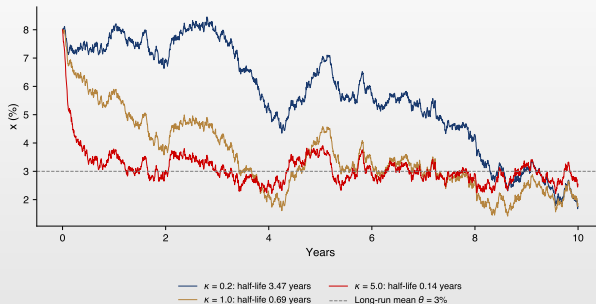
Solution and Stationary Distribution

- By Itô's lemma on $e^{\kappa t} x_t$: $x_t = \theta + (x_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s$
- Conditional distribution: Normal with
 - ▶ $\mathbb{E}[x_t | x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$, $\text{Var}(x_t | x_0) = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa t})$
- Stationary distribution ($t \rightarrow \infty$): $N(\theta, \sigma^2/(2\kappa))$
- **Half-life** $\ln 2/\kappa$: the time for an expected deviation from θ to halve
 - ▶ $\kappa = 0.5$ per year: half-life 1.39 years

Exact Discretisation: an AR(1)

- Sampled at step Δt the OU process is exactly an autoregressive model of order 1 (AR(1)):
 - ▶ $x_{t+\Delta t} = a + b x_t + \varepsilon_t$, $b = e^{-\kappa \Delta t}$, $a = \theta(1 - b)$, $\text{Var}(\varepsilon_t) = \frac{\sigma^2}{2\kappa}(1 - b^2)$
- Maximum likelihood = ordinary least squares (OLS) of $x_{t+\Delta t}$ on x_t , then invert the mapping
 - ▶ $\hat{\kappa} = -\ln \hat{b} / \Delta t$, $\hat{\theta} = \hat{a} / (1 - \hat{b})$, $\hat{\sigma} = \hat{s}_\varepsilon \sqrt{2\hat{\kappa} / (1 - \hat{b}^2)}$
- Standard errors for $\hat{\kappa}$ and $\hat{\theta}$ by the delta method from the OLS covariance of $(\hat{a}, \hat{b})$
- With daily data b is very close to 1: κ is hard to estimate, as with unit-root tests

Three Speeds of Mean Reversion



- OU paths driven by the same shocks, starting at 8%, with $\theta = 3\%$, $\sigma = 1\%$ and $\kappa = 0.2, 1, 5$; exact simulation, daily steps

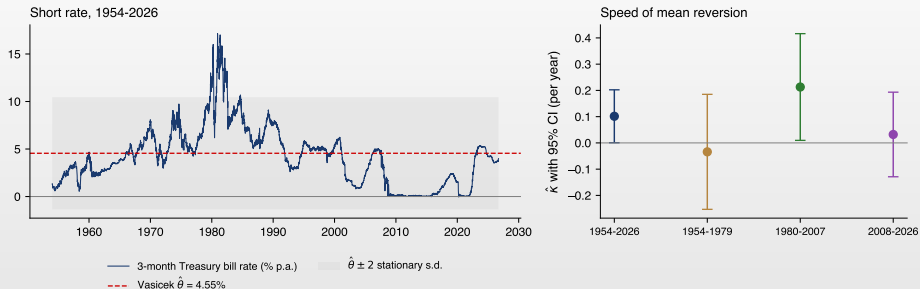
Reading the OU Chart

- Half-lives: 3.47, 0.69 and 0.14 years
 - ▶ With $\kappa = 5$ the gap from 8% to 3% closes within a few months
- Stationary standard deviations $\sigma/\sqrt{2\kappa}$: 1.58%, 0.71%, 0.32%
 - ▶ Fast reversion keeps the process in a narrow band; slow reversion looks like a random walk for years
- Unlike Brownian motion, the variance stops growing: long-horizon forecasts converge to θ

Short-Rate Models: Vasicek and CIR

- **Vasicek**: the short rate r_t is an OU process [Vasicek \(1977\)](#)
 - ▶ Normal distribution: negative rates have positive probability $\Phi(-\theta\sqrt{2\kappa}/\sigma)$ in the long run
 - ▶ Bond prices and yields in closed form (Feynman–Kac, Section 8)
- **Cox–Ingersoll–Ross (CIR)**: $dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dW_t$ [Cox, Ingersoll & Ross \(1985\)](#)
 - ▶ Volatility shrinks near zero; the rate stays positive if $2\kappa\theta \geq \sigma^2$ (Feller condition) [Feller \(1951\)](#)
- Which is closer to the data is an empirical question, tested in Section 7 [Chan, Karolyi, Longstaff & Sanders \(1992\)](#)

Vasicek on the 3-Month Treasury Bill Rate



- Left: daily 3-month US Treasury bill rate (DTB3), 1954–2026, with the Vasicek long-run mean; right: $\hat{\kappa}$ with 95% CI by period

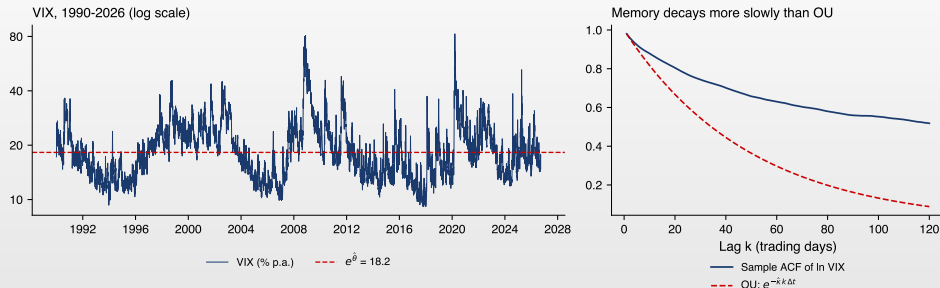
Reading the Vasicek Chart

- ▣ Full sample: $\hat{\theta} = 4.55\%$, $\hat{\kappa} = 0.101$ (s.e. 0.051), half-life 6.8 years, $\hat{\sigma} = 1.33\%$
 - ▶ Daily AR(1) coefficient $\hat{b} = 0.99960$: almost a unit root
- ▣ Subperiods disagree: $\hat{\kappa} = -0.034$ (1954–1979, explosive), 0.213 (1980–2007), 0.032 (2008–2026)
 - ▶ Two of the three 95% CIs include zero: mean reversion is not detectable within 20–25 years
- ▣ The rate ranged from -0.05% (2020-03-26) to 17.1% (1980-12-11); 4.3% of days were below 0.05%
 - ▶ The fitted Vasicek model gives a 6.1% long-run probability of a negative rate

The Mean-Reversion Speed Is Biased Upwards

- In finite samples $\hat{\kappa}$ overstates κ ; the bias is roughly $4/T$ with T in years [Tang & Chen \(2009\)](#)
 - ▶ It depends on the calendar span, not on the sampling frequency, like the drift of GBM
- Simulation with the full-sample Vasicek estimates as the truth (72 years, 500 samples)
 - ▶ Mean $\hat{\kappa} = 0.164$ against the true 0.101: bias 0.063, close to $4/T = 0.055$
 - ▶ 78% of the estimates exceed the truth
- Consequence: estimated half-lives are too short, and mean reversion looks faster than it is

Is Volatility Mean-Reverting? The VIX



- Left: VIX 1990–2026 on a log scale with the OU long-run level of $\ln VIX$; right: sample ACF of $\ln VIX$ against the OU-implied $e^{-\hat{\kappa} k \Delta t}$

Reading the VIX Chart

- OU fit to $\ln VIX$: $\hat{\kappa} = 5.10$ (s.e. 0.53), half-life 34 trading days, long-run level $e^{\hat{\theta}} = 18.2$
 - ▶ Volatility spikes (VIX 82.7 on 2020-03-16) fade within weeks, not years
- But the ACF at lag 60 is 0.63, against 0.30 implied by OU
 - ▶ Fast decay first, then a slow component: one OU factor is too simple
 - ▶ Candidates: two-factor or long-memory models; rough volatility [Gatheral, Jaisson & Rosenbaum \(2018\)](#) is a distinct alternative without long memory; which fits better must be tested
- Mean-reverting volatility is the building block of the Heston model

Recap: Mean Reversion

- OU: a linear pull towards θ with speed κ ; Normal, stationary, exactly an AR(1) when sampled
- Half-life $\ln 2 / \kappa$ summarises the persistence
- Interest rates: κ small, unstable across periods and biased upwards in finite samples
- Volatility (VIX): clear mean reversion within weeks, plus a slow component

Infill versus Long-Span Asymptotics

- **Infill:** $\Delta \rightarrow 0$ on a fixed $[0, T]$; then $\sum_k (\Delta X_k)^2 \rightarrow \int_0^T b_t^2 dt$
 - ▶ The diffusion coefficient is identified **without error** in the limit: quadratic variation is observable
- The drift is **not** consistently estimable on a fixed span: its Fisher information grows with T , not with n
 - ▶ GBM: $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$ (Section 5); OU: $\text{Var}(\hat{\kappa}) \approx (e^{2\kappa\Delta} - 1)/(T\Delta) \rightarrow 2\kappa/T$ as $\Delta \rightarrow 0$ (Seminar A9)
- **Long span** ($T \rightarrow \infty$) is needed for μ, κ, θ ; the bias of $\hat{\kappa}$ is not reduced by finer sampling [Yu \(2012\)](#)
 - ▶ $b = e^{-\kappa\Delta} \rightarrow 1$: a near unit root, so $\hat{\kappa}$ is far from Normal when κT is small
- One phenomenon, not two facts: the imprecise GBM drift and the biased $\hat{\kappa}$ are both long-span problems

Likelihood for Nonlinear Diffusions

- **Euler pseudo-likelihood:** a Gaussian transition $r_{t+\Delta} \mid r_t \sim N(r_t + a(r_t)\Delta, b^2(r_t)\Delta)$
 - ▶ Biased at order Δ : small with daily data, larger with monthly or quarterly steps
- **Exact CIR:** $2c r_{t+\Delta} \mid r_t$ is noncentral χ^2 with $4\kappa\theta/\sigma^2$ degrees of freedom [Cox, Ingersoll & Ross \(1985\)](#)
 - ▶ $c = 2\kappa/(\sigma^2(1 - e^{-\kappa\Delta}))$, noncentrality $2c r_t e^{-\kappa\Delta}$; Feller condition $\Leftrightarrow$ degrees of freedom ≥ 2 (Seminar A5)
- General diffusions: closed-form Hermite expansions of the transition density [Aït-Sahalia \(2002\)](#)
- CKLS (Chan–Karolyi–Longstaff–Sanders): $dr = \kappa(\theta - r)dt + \sigma r^\gamma dW$ nests Vasicek ($\gamma = 0$) and CIR ($\gamma = 1/2$) [Chan, Karolyi, Longstaff & Sanders \(1992\)](#)
 - ▶ Gaussian (Euler) quasi-likelihood; tests on γ with sandwich standard errors whose score covariance is heteroskedasticity- and autocorrelation-consistent (HAC) [Newey & West \(1987\)](#)

Vasicek, CIR and CKLS on the T-Bill Rate, 1954–2007

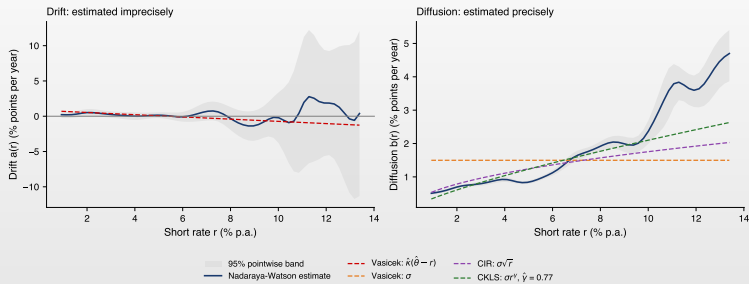
Model	$\hat{\kappa}$ daily (s.e.)	$\hat{\kappa}$ monthly (s.e.)	γ
Vasicek, exact	0.158 (0.073)	0.201 (0.083)	0
Vasicek, Euler	0.158 (0.073)	0.199 (0.082)	0
CIR, exact	0.113 (0.060)	0.153 (0.070)	1/2
CIR, Euler	0.093 (0.060)	0.144 (0.069)	1/2
CKLS, Euler	0.080 (0.057)	0.144 (0.068)	0.77 (0.033); 0.60 (0.149)

- Daily DTB3 ($n = 13\,487$) and month-end values ($n = 647$); 2008–2026 excluded: 13 exact zeros and 8 negative quotes lie outside the domain of these likelihoods, and 17% of the days are a near-zero regime below 0.05%
- Exact and Euler differ by less than one standard error: discretisation error is small next to the long-span uncertainty of $\hat{\kappa}$
- HAC Wald tests (11 and 6 lags): $\gamma = 0$ rejected (daily 545, monthly 16); $\gamma = 1/2$ rejected daily (67.0; 22.1 with 55 lags), not monthly (0.5 against 3.84)
- The Gaussian LR for $\gamma = 1/2$ rejects monthly (10.1): it ignores volatility clustering; [Chan, Karolyi, Longstaff & Sanders \(1992\)](#) found $\hat{\gamma} \approx 1.5$ for 1964–1989

Nonparametric Drift and Diffusion

- No functional form: kernel averages of the increments around each level x
 - ▶ $\hat{a}(x) = \sum_t w_t(x) \Delta r_t / \Delta$, $\hat{b}^2(x) = \sum_t w_t(x) (\Delta r_t)^2 / \Delta$, $w_t(x) \propto K((r_t - x)/h)$
 - ▶ Nadaraya–Watson weights, Gaussian kernel K , bandwidth h ; $\mathbb{E}[\Delta r \mid r = x] / \Delta = a(x) + O(\Delta)$ and $\mathbb{E}[(\Delta r)^2 \mid r = x] / \Delta = b^2(x) + O(\Delta)$: exact only as $\Delta \rightarrow 0$
- Stanton (1997): nonparametric drift, diffusion and market price of interest-rate risk
- Bandi & Phillips (2003): valid for non-stationary but recurrent processes, which matters because the T-bill rate is close to a unit root
- The diffusion converges at the infill rate, the drift only at the long-span rate
- Caution: the nonparametric specification test of Aït-Sahalia (1996) rejects far too often with persistent data Pritsker (1998)

Nonparametric Drift and Diffusion of the T-Bill Rate



- Daily DTB3, 1954–2007; bandwidth $h = 0.44$ percentage points; 95% heteroskedasticity-robust pointwise bands (serial dependence ignored)
- Drift: zero lies inside the band at 97% of the levels; diffusion: median relative s.e. 8%, $b(r)$ rises from 0.5% to 4.7% (log-slope 0.90)
- No parametric shape fits everywhere: inside the band at 35% (CKLS), 7% (CIR), 5% (Vasicek) of the levels

Recap: Estimating Diffusions

- Infill identifies the diffusion coefficient; the drift, κ and θ need a long calendar span
- Euler versus exact likelihood: the discretisation bias is small next to the sampling error of the drift
- Short-rate volatility rises with the level: Vasicek is rejected; γ lies between 1/2 and 1 depending on the frequency
- Nonparametric estimates: precise diffusion, imprecise drift; specification tests on persistent data over-reject

Girsanov's Theorem

- Real-world measure P and an equivalent measure Q (the same events of probability zero), linked by a **Radon–Nikodym density**
 - ▶ $\frac{dQ}{dP} \Big|_{\mathcal{F}_T} = Z_T = \exp \left(- \int_0^T \theta_s dW_s - \frac{1}{2} \int_0^T \theta_s^2 ds \right)$
- **Girsanov (1960)**: under Q , $\tilde{W}_t = W_t + \int_0^t \theta_s ds$ is a Brownian motion if $\mathbb{E} \left[e^{\frac{1}{2} \int_0^T \theta_s^2 ds} \right] < \infty$ (Novikov condition)
- For $dX = a dt + b dW$: $dX = (a - b\theta) dt + b d\tilde{W}$
 - ▶ **Only the drift changes**: the diffusion coefficient and the quadratic variation are the same under P and Q
 - ▶ Under GBM (constant σ), realised variance (Chapter 9) therefore measures the σ that prices options, while μ drops out
 - ▶ With stochastic volatility, P and Q weigh future variance paths differently: historical variance is a benchmark, not a calibration (next slide)
- Textbook treatment: [Shreve \(2004\)](#), Chapters 5–6; [Franke, Härdle & Hafner \(2019\)](#)

Market Price of Risk: P versus Q

- GBM: $\theta = (\mu - r)/\sigma$ is the **market price of risk**; under Q , $dS = rS dt + \sigma S d\tilde{W}$ (Seminar A1)
 - ▶ S&P 500, 1990–2026: $\hat{\theta} = (10.0\% - 2.7\%)/18.0\% = 0.40$, s.e. $1/\sqrt{T} = 0.17$ (r : mean 3-month T-bill rate)
 - ▶ The discounted price $e^{-rt}S_t$ is a Q -martingale
- Vasicek with risk premium $\lambda\sigma$: $\kappa^Q = \kappa$, $\kappa\theta^Q = \kappa\theta - \lambda\sigma$
 - ▶ The long-run level that prices bonds differs from the one estimated on data (Section 6)
- Heston with volatility risk premium λv_t : $\kappa^Q = \kappa + \lambda$, $\kappa^Q\theta^Q = \kappa\theta$
 - ▶ $\lambda < 0$ (investors pay for protection against variance) gives $\theta^Q > \theta$: the variance risk premium of Chapter 9

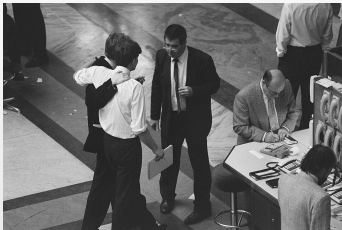
Feynman–Kac: From Expectations to the Pricing PDE

- If $dX = a(X) dt + b(X) d\tilde{W}$ under Q Kac (1949)
 - ▶ $u(t, x) = \mathbb{E}^Q \left[e^{-\int_t^T r_s ds} g(X_T) \mid X_t = x \right]$ solves $u_t + a u_x + \frac{1}{2} b^2 u_{xx} - r u = 0$, $u(T, x) = g(x)$
- Proof idea: $e^{-\int_0^t r_s ds} u(t, X_t)$ is a Q -martingale, so Itô's lemma must give it a zero drift
- Vasicek bond ($g = 1$, $X = r$): $P(t, T) = \exp(A(\tau) - B(\tau) r_t)$, $\tau = T - t$ (Seminar A3)
 - ▶ $B(\tau) = \frac{1 - e^{-\kappa \tau}}{\kappa}$, $A(\tau) = \left(\theta^Q - \frac{\sigma^2}{2\kappa^2} \right) (B(\tau) - \tau) - \frac{\sigma^2 B(\tau)^2}{4\kappa}$
 - ▶ Section 6 estimates with $\lambda = 0$ and $r = 3.99\%$: 10-year yield 4.05%; the 10-year Treasury yield (FRED DGS10) on 2026-09-18: 5.01%
- The gap is a term premium ($\lambda \neq 0$) plus misspecification; Chapter 12 derives Black–Scholes by the same route

Recap: Change of Measure

- Girsanov: an equivalent change of measure shifts the drift by $-b\theta$ and leaves the volatility unchanged
- The market price of risk links P and Q : drift parameters differ across measures, diffusion parameters do not
- Feynman–Kac turns a Q -expectation into a pricing PDE: bond prices here, option prices in Chapter 12
- The market price of risk for the S&P 500 is estimated with a standard error of 0.17 from 37 years: as imprecise as the drift

Why Jumps? Poisson and Compound Poisson



Amsterdam Stock Exchange, 21 October 1987, after
the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia
Commons

- 19 October 1987: the Dow Jones fell 22.6% in one day, about 20 standard deviations under GBM with 18% volatility
- Brownian paths are continuous; real prices gap on news, and a jump cannot be hedged by continuous trading [Merton \(1976\)](#)
- N_t Poisson with intensity λ ; **compound Poisson** $J_t = \sum_{i \leq N_t} Y_i$, sizes Y_i i.i.d., independent of N
 - Cumulants $\kappa_k(J_t) = \lambda t \mathbb{E}[Y^k]$; Merton: $Y_i \sim N(\mu_J, \sigma_J^2)$, jumps in the **log** price

Merton's Jump-Diffusion

- Log price = Brownian motion with drift + compound Poisson jumps [Merton \(1976\)](#)
 - ▶ $d \ln S_t = m dt + \sigma dW_t + dJ_t$, $J_t = \sum_{i \leq N_t} Y_i$
- Return over Δt given k jumps: $N(m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$
 - ▶ Unconditional density: a **Poisson mixture** of Normal densities
 - ▶ $f(r) = \sum_{k \geq 0} e^{-\lambda\Delta t} \frac{(\lambda\Delta t)^k}{k!} \varphi(r; m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$
- Five parameters ($m, \sigma, \lambda, \mu_J, \sigma_J$); returns still i.i.d. over time
- Exact simulation: draw $N \sim \text{Poisson}(\lambda\Delta t)$, then a Normal with the conditional moments

Moments of the Merton Return

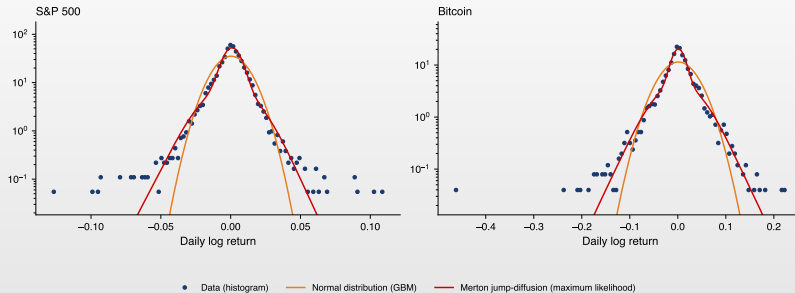
- Cumulants of the return over Δt (the k -th cumulant of a compound Poisson sum is $\lambda \Delta t \mathbb{E}[Y^k]$):
 - ▶ $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2)$, $\kappa_3 = \lambda \Delta t (\mu_J^3 + 3\mu_J \sigma_J^2)$
 - ▶ $\kappa_4 = \lambda \Delta t (\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$; skewness $\kappa_3 / \kappa_2^{3/2}$, excess kurtosis κ_4 / κ_2^2
- Negative μ_J gives negative skewness; any jump gives positive excess kurtosis
- Excess kurtosis falls like $1/\Delta t$ as the horizon grows: jumps matter most at short horizons
 - ▶ Example: $\sigma = 15\%$, $\lambda = 5$, $\mu_J = -3\%$, $\sigma_J = 4\%$: daily excess kurtosis 17.6, skewness -2.07

Maximum Likelihood Estimates

	$\hat{\sigma}$ (%)	$\hat{\lambda}$ (per year)	$\hat{\mu}_J$ (%)	$\hat{\sigma}_J$ (%)	Jump share of variance	LR
S&P 500	9.4	102 (11)	-0.16	1.44	71%	2 401
BET	11.3	79 (7)	-0.09	2.07	73%	2 326
Bitcoin	24.6	239 (21)	0.00	3.85	85%	1 357

- Log-likelihood maximised numerically over the Poisson mixture (up to 12 jumps per day), standard errors from the Hessian in brackets
- LR against GBM ($\lambda = 0$) is not χ^2_3 : $\lambda = 0$ is on the boundary and μ_J, σ_J are unidentified; bootstrap critical values in Seminar B6
- The fit prefers **many small** jumps (about 80–240 per year), not a few crashes

Merton Fit to the Return Distribution



- Density of daily log returns on a log scale: histogram, the Normal distribution implied by GBM and the fitted Merton mixture

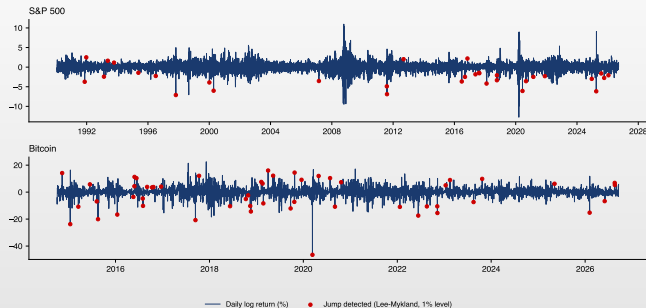
Reading the Merton Fit

- The Merton density follows the peak and the shoulders far better than the Normal distribution
 - ▶ LR statistics in the thousands, far beyond the bootstrap critical value: GBM is rejected
- S&P 500: implied excess kurtosis 3.7 against 10.9 in the data; skewness -0.31 against -0.36
 - ▶ The most extreme days lie beyond both fitted curves
- About 102 “jumps” per year means the model uses jumps to mimic **changing volatility**
 - ▶ An i.i.d. model cannot tell a jump from a volatile period: we need a direct test for jumps
- The likelihood is **unbounded**: with $m\Delta t$ equal to one observed return and $\sigma \rightarrow 0$, that density explodes; the MLE is a local maximum, and the LR test is defined on $\sigma \geq 5\%$ (Seminar B6) [Honoré \(1998\)](#)

Detecting Jumps: the Lee–Mykland Test

- Idea: a jump is a return that is huge **relative to the local volatility** Lee & Mykland (2008)
 - ▶ $L_t = r_t / \hat{\sigma}_t$, $\hat{\sigma}_t^2 = \frac{1}{K-2} \sum_{j=t-K+2}^{t-1} |r_j| |r_{j-1}|$ (bipower variation, robust to jumps)
- Under no jumps, the maximum of $|L_t|$ over n days has a Gumbel limit
 - ▶ Reject if $(|L_t| - C_n) / S_n > -\ln(-\ln(1 - \alpha))$; at $\alpha = 1\%$ this threshold for the **normalised** statistic, not for $|L_t|$, is 4.60
 - ▶ $C_n = \frac{\sqrt{2 \ln n}}{c} - \frac{\ln \pi + \ln \ln n}{2c\sqrt{2 \ln n}}$, $S_n = \frac{1}{c\sqrt{2 \ln n}}$, $c = \sqrt{2/\pi}$ (the mean of $|Z|$, which rescales the bipower estimate), n = number of tested days
- Daily data: window $K = 16$ days, as recommended by the authors
- Controls, asymptotically, the probability of a false jump over the **whole** sample, not day by day

Jumps Detected in the S&P 500 and Bitcoin



□ Daily log returns (%) with the days flagged by the Lee–Mykland test at the 1% level

Reading the Jump Chart

- ▣ S&P 500: 32 jumps in 36.7 years, 0.87 per year; 84% of them are falls
 - ▶ Not one of the 10 largest absolute returns is flagged: they came in turbulent periods (2008, 2020)
 - ▶ Instead, moves as small as 1.2% in a very calm market are jumps (1993-10-14)
- ▣ Bitcoin: 52 jumps, 4.35 per year, 5 of its 10 largest moves flagged
 - ▶ Up and down jumps equally often (52% negative)
- ▣ The test flags only a few candidate jumps per year, far fewer than the Merton fit; the gap may reflect stochastic volatility, drift bursts and limited detection power in daily data

Recap: Jumps

- Merton adds compound Poisson jumps to GBM; returns become a Poisson mixture of Normal distributions
- Jumps create skewness and excess kurtosis, strongest at short horizons
- Fitted by MLE on i.i.d. returns, the model mistakes volatile periods for frequent jumps
- Nonparametric tests relative to local volatility flag only a few candidate jumps per year in daily data

The Heston Model

- Two SDEs, for the log price and for the instantaneous variance v_t [Heston \(1993\)](#)
 - ▶ $d \ln S_t = (\mu - \frac{1}{2} v_t) dt + \sqrt{v_t} dW_t^{(1)}$
 - ▶ $dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$
- Variance follows a CIR process: mean-reverting to θ with speed κ ; ξ = volatility of volatility
 - ▶ Feller condition $2\kappa\theta \geq \xi^2$: variance never touches zero [Feller \(1951\)](#)
- $\rho < 0$: **leverage effect**, falling prices come with rising volatility
 - ▶ Simulation: Euler scheme with full truncation $v^+ = \max(v, 0)$ [Lord, Koekkoek & van Dijk \(2010\)](#); an alternative discretisation, the quadratic-exponential scheme [Andersen \(2008\)](#)

Heston Parameters from the VIX

- Proxy $y_t = (\text{VIX}_t/100)^2$ in place of v_t ; discretised CIR regression
 - ▶ $\frac{y_{t+\Delta t} - y_t}{\sqrt{y_t}} = \kappa\theta \frac{\Delta t}{\sqrt{y_t}} - \kappa\Delta t\sqrt{y_t} + \xi\sqrt{\Delta t}\varepsilon_t$; ρ = correlation of return and variance shocks
- S&P 500 and VIX, 1990–2026: $\hat{\kappa} = 4.13$ (half-life 42 days), $\hat{\xi} = 0.56$, $\hat{\rho} = -0.69$
 - ▶ $\sqrt{\hat{\theta}} = 20.9\%$ from the VIX, but the realised volatility is 18.0%; Feller ratio of the proxy $2\hat{\kappa}\hat{\theta}/\hat{\xi}^2 = 1.16$
- But y_t is **not** v_t : it is a risk-neutral expectation of future variance, affine in v_t (next slide)
 - ▶ For real-world paths we set θ to the realised variance; $\hat{\xi}$ is a lower bound for ξ

What the VIX Identifies in Heston

□ Under Q : $y_t = \frac{1}{\tau} \mathbb{E}^Q \left[\int_t^{t+\tau} v_s ds \mid \mathcal{F}_t \right] = a + b v_t$, $\tau = 30/365$ (Seminar A6)

▶ $b = \frac{1 - e^{-\kappa^Q \tau}}{\kappa^Q \tau} < 1$, $a = \theta^Q(1 - b)$: affine in v_t , with Q -parameters

□ What the regression on y_t recovers

▶ κ : the P -speed, since an affine map keeps the dynamics of v_t ; ρ : unchanged, correlation is scale-free

▶ ξ : attenuated; the diffusion of y is $\xi \sqrt{b(y - a)}$, so $\xi_y = b \xi \sqrt{v/(a + bv)}$ varies with the state and equals $b \xi$ only at $v_t = \theta^Q$

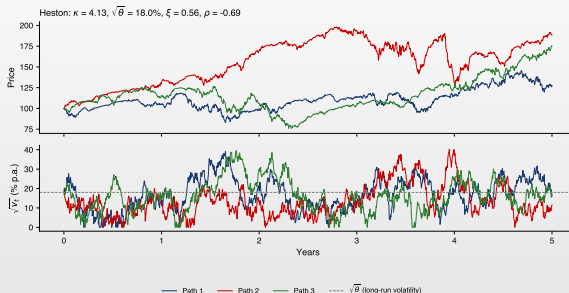
▶ The long-run level $a + b \theta^P$ mixes θ^Q and θ^P : recover $\theta^P = (\theta_y - a)/b$ before a Feller check on v_t

κ^Q	1	2,5	5	10
b	0.96	0.90	0.82	0.68
Local correction $\hat{\xi}/b$	0.58	0.62	0.68	0.82

□ $\hat{\xi}/b$ is exact only at $v_t = \theta^Q$; proper estimation fits the shifted square-root dynamics of y , or v_t and the observation equation jointly

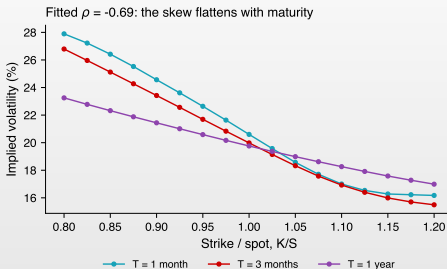
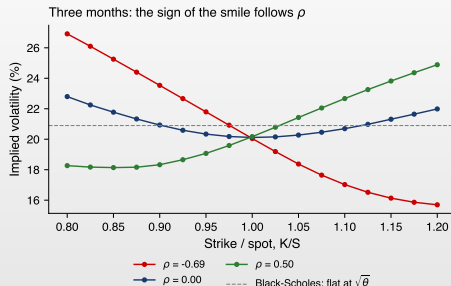
□ κ^Q is not identified from the VIX alone: joint P/Q estimation with option prices [Chernov & Ghysels \(2000\)](#), or moments of realised variance [Bollerslev & Zhou \(2002\)](#)

Simulated Heston Paths



- Three five-year paths with the VIX-based $\hat{\kappa}$, $\hat{\xi}$ (uncorrected), $\hat{\rho}$ and θ equal to the realised S&P 500 variance; top: price, bottom: $\sqrt{V_t}$
- Clustering now comes from the model; price falls coincide with volatility spikes (simulated return–variance correlation -0.69 , the leverage effect through ρ)
- With θ at the realised variance the Feller ratio is $2\hat{\kappa}\theta/\hat{\xi}^2 = 0.86 < 1$: the CIR variance can reach zero; negative values arise only in the Euler approximation, and full truncation sets them to zero (1.4% of the steps)

A Preview of Chapter 12: the Volatility Smile



- Black-Scholes implied volatilities of Heston call prices (Monte Carlo, 100,000 antithetic paths); left: effect of ρ at 3 months; right: maturities with the fitted ρ

Reading the Smile Chart

- ▣ **Implied volatility**: the σ that makes the Black–Scholes price equal to a given option price
 - ▶ Under GBM it is the same for every strike: a flat line
- ▣ With $\hat{\rho} = -0.69$, three months: 26.9% at strike 80% of spot, 20.0% at the money, 15.7% at 120%
 - ▶ A downward **skew**, as in equity-index options; with $\rho = 0$ a symmetric **smile** (22.8%, 20.1%, 22.0%)
- ▣ The skew flattens with maturity: from 27.9% – 16.2% at one month to 23.2% – 17.0% at one year
- ▣ Illustration only: the VIX-regression estimates are used as Q -parameters; pricing needs κ^Q , θ^Q calibrated to option prices, as in Chapter 12

Affine Jump-Diffusions

- A model is **affine** if drift, variance and jump intensity are affine functions of the state X_t [Duffie, Pan & Singleton \(2000\)](#)
 - ▶ Vasicek, CIR, Merton, Heston and Bates are all affine
- Then the conditional characteristic function is exponential-affine in the state
 - ▶ $\mathbb{E} \left[e^{iu^\top X_\tau} \mid X_t \right] = \exp \left(\alpha(\tau) + \beta(\tau)^\top X_t \right)$, with α, β solving Riccati ordinary differential equations
- This is why Heston has “only the characteristic function” (Section 4): option prices follow by Fourier inversion
- The Vasicek bond of Section 8 is the simplest case: $A(\tau)$ and $B(\tau)$ solve Riccati equations

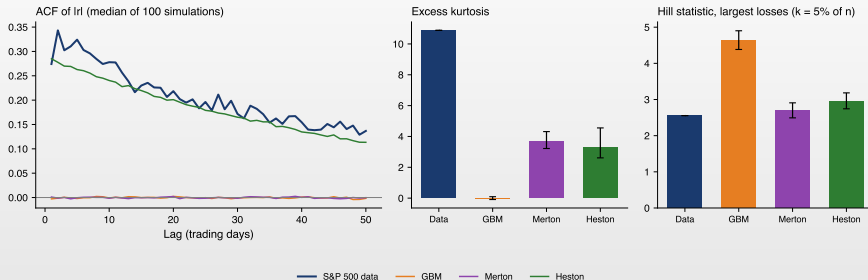
From GARCH to Continuous Time

- Nelson (1990): as the time step shrinks, GARCH(1,1) converges to a continuous-time stochastic-volatility diffusion
 - ▶ $dv_t = \kappa(\theta - v_t)dt + \xi v_t dW_t^{(2)}$: like Heston, but with v_t instead of $\sqrt{v_t}$
- The GARCH persistence $\alpha + \beta$ of Chapter 5 maps to $\kappa \approx (1 - \alpha - \beta)/\Delta t$
 - ▶ $\alpha + \beta = 0.99$ with daily data gives $\kappa \approx 2.5$ per year, a half-life of about 70 days
- Discrete-time GARCH for forecasting and risk; continuous-time stochastic volatility for pricing and hedging
- Caution: GARCH and its limit are **not** asymptotically equivalent experiments (one noise source against two): inference can differ even as $\Delta t \rightarrow 0$ Wang (2002)
- Further reading: Čížek, Härdle & Weron (2011) (Heston model chapter), Härdle, Chen & Overbeck (2017)

Recap: Stochastic Volatility

- ▣ Heston: variance is a mean-reverting CIR process correlated with returns
- ▣ Parameters can be read from the VIX: fast mean reversion, high volatility of volatility, strong leverage
- ▣ Generates volatility clustering and, through ρ , the implied-volatility skew
- ▣ GARCH is its discrete-time cousin

Which Model Reproduces the S&P 500?



- 100 simulated samples of 9 245 days from each fitted model: median ACF of $|r_t|$, excess kurtosis and the Hill statistic of the largest losses, with 90% ranges

Reading the Model Comparison

	Data	GBM	Merton	Heston
Excess kurtosis	10.88	0.00	3.66	3.30
Skewness	-0.36	0.00	-0.32	0.00
Hill statistic	2.56	4.64	2.70	2.95
ACF of $ r $, lag 1	0.27	0.00	0.00	0.28
ACF of $ r $, sum of lags 1–20	5.32	0.00	0.00	4.78

- Medians of 100 simulations; Hill statistic of the largest losses with $k = 5\%$ of all returns (lower = heavier tail): a tail index only under regular (Pareto-type) variation, descriptive otherwise; data: 2.97, 2.56, 2.04 for $k = 2.5\%$, 5% , 10%
- These fitted specifications: Merton gets the Hill statistic and skewness but no clustering; Heston, with the VIX-based parameters, gets the clustering but no skewness
- No fitted model reaches the kurtosis of 10.9; a Heston model with a lower Feller ratio could (short-horizon excess kurtosis $\approx 3\xi^2/(2\kappa\theta)$), but the evidence of [Andersen, Benzoni & Lund \(2002\)](#) and [Eraker, Johannes & Polson \(2003\)](#) favours jumps and stochastic volatility (SVJ, [Bates \(1996\)](#))

Which Model When?

Model	Captures	Typical use
Geometric Brownian motion	Positive prices, $\sqrt{t}$ scaling	Black–Scholes benchmark, long-horizon planning
Ornstein–Uhlenbeck, CIR	Mean reversion	Interest rates, spreads, volatility indices
Merton jump-diffusion	Gaps, short-horizon fat tails	Short-dated options, credit events, crypto
Heston stochastic volatility	Clustering, leverage, skew	Equity-index options, volatility trading
Bates (Heston + jumps)	All of the above	Option surfaces across maturities

Beyond This Chapter

- ▣ Heavier tails: Lévy processes and α -stable distributions [Pele \(2014\)](#)
 - ▶ Crypto-assets form a distinct class by the shape of their return distributions [Pele et al. \(2023\)](#)
- ▣ Rough volatility: volatility paths rougher than Brownian motion [Gatheral, Jaisson & Rosenbaum \(2018\)](#)
 - ▶ High-frequency information about Bitcoin volatility and tail risk [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- ▣ Exercises with solutions on all models of this chapter: [Borak, Härdle & López-Cabrera \(2013\)](#)
- ▣ Next, Chapter 12: pricing options by replication and reading the volatility surface

Research Question: Rough, Yet Persistent?

- Paper: [Bennedsen, Lunde & Pakkanen \(2022\)](#); section, table and figure numbers follow the accepted version [arXiv:1610.00332v3](#)
 - ▶ It builds on [Gatheral, Jaisson & Rosenbaum \(2018\)](#): log-volatility as a fractional OU process with Hurst exponent $H < 1/2$ (RFSV), whereas the long-memory model of [Comte & Renault \(1998\)](#) uses $H > 1/2$ (Section 1)
- Roughness index α : $1 - \rho(h) \sim c|h|^{2\alpha+1}$ as $h \rightarrow 0$, with ρ the autocorrelation of log-volatility (eq. (2.1)); $\alpha < 0$: paths rougher than Brownian motion
 - ▶ Memory parameter β : $\rho(h) \sim c|h|^{-\beta}$ as $h \rightarrow \infty$ (eq. (2.2)); $\beta < 1$: the ACF is not integrable, i.e. long memory
- The problem: in the fractional OU process a single H fixes both, $\alpha = H - 1/2$ and $\beta = 2 - 2H$ (Section 2): with $H < 1/2$, $\beta > 1$, so a rough model cannot have long memory
 - ▶ Research question: are realised volatilities both rough and persistent, and do models that set α and β separately forecast better? (Sections 1, 5 and 6)
- Link to this chapter: the stationary OU process $\int_{-\infty}^t e^{-\lambda(t-s)} dW_s$ is the special case $g(x) = e^{-\lambda x}$, $\alpha = 0$, of the kernel model on the next slide

Case Study: Data and Method

- **Model:** $\sigma_t = \xi e^{X_t}$ (eq. (2.3)), $X_t = \int_{-\infty}^t g(t-s) v_s dW_s$, a Brownian semistationary (BSS) process with kernel g (eq. (2.5))
 - ▶ Gamma kernel $g(x) = x^\alpha e^{-\lambda x}$ (Example 2.2): roughness α , short memory with exponential decay at rate λ
 - ▶ Power kernel $g(x) = x^\alpha (1+x)^{-\gamma-\alpha}$ (eq. (2.8)): $\beta = \gamma$ if $\gamma > 1$, $\beta = 2\gamma - 1$ if $\gamma \in (1/2, 1)$ (Remark 2.2); Gaussian alternative: the Cauchy class, $\rho(h) = (1 + |h|^{2\alpha+1})^{-\beta/(2\alpha+1)}$ (Section 2.1.1)
- **Step 1,** α from the variogram $\hat{\gamma}_2(k)$ of $\log \hat{\sigma}$: OLS of $\log \hat{\gamma}_2(k)$ on $\log k$, $k = 1, \dots, 6$, $\hat{\alpha} = (\hat{\alpha}_1 - 1)/2$ (eq. (3.1))
 - ▶ Noise-robust NLLS: $\hat{\gamma}_2(k) = b_0 + b_1(k\Delta)^{2\alpha+1}$, the constant absorbs the measurement noise; average over $m = 10, \dots, 20$ lags (Sections 3.1 and 4.1)
 - ▶ **Step 2:** β , γ or λ by matching the model ACF to the empirical one on the first $\lceil n^{1/3} \rceil$ lags (Sections 3.2 and 5.1)
- **Data:** E-mini S&P 500 futures, every trade, 2011–2014 (996 days), 09:30–16:00 New York time; $\hat{\sigma}_t^2 =$ pre-averaged bipower variation over Δ , divided by Δ , for Δ from 10 minutes to 1 day (Sections 5 and 5.1)
 - ▶ Intraday seasonality removed with the flexible Fourier form of [Andersen & Bollerslev \(1997\)](#); daily data on 1944 liquid US stocks, 2003–2013 (Section 5.2)
- **Forecasts:** variance summed over the next $h = 1, 2, 5, 10, 20$ steps, rolling window of 200 observations, out of sample 2 January 2013 – 31 December 2014 (Section 6.1)
 - ▶ Rivals: random walk, AR(5), AR(10), ARMA(1,1), log-HAR [Corsi \(2009\)](#), ARFIMA(0, d ,0), RFSV; losses MSE and QLIKE; 90% and 75% MCS [Hansen, Lunde & Nason \(2011\)](#)

Case Study: Results

$\log \sigma$, E-mini S&P 500 (Table 3, Panel A)	$\hat{\alpha}_{OLS}$	$\hat{\alpha}_{NLLS}^*$	$\hat{\beta}_{Cauchy}^*$	$\hat{\lambda}_{Gamma}^*$
$\Delta = 10$ minutes	-0.38	-0.38	0.18	0.02
$\Delta = 65$ minutes	-0.30	-0.35	0.18	0.02
$\Delta = 1$ day	-0.30	-0.33	0.18	0.02

- Rough at every scale: $\hat{\alpha}$ between -0.38 and -0.30 from 10 minutes to 1 day (Table 3, Panel A; * = noise-robust): log-volatility is far rougher than Brownian motion ($\alpha = 0$)
 - ▶ And persistent: $\hat{\beta}_{Cauchy}^*$ between 0.17 and 0.18 at every Δ , below 1, and $\hat{\lambda}^*$ between 0.02 and 0.03 (Table 3, Panel A): long memory together with roughness
 - ▶ 1 944 US stocks: $\hat{\alpha}^*$ mostly between -0.45 and -0.30 , $\hat{\beta}^*$ mostly between 0 and 0.4, no systematic difference across sectors (Figures 4 and 5): both features look universal
- Noise can fake roughness: a simulated $\alpha = 0$ observed with noise-to-signal ratio 0.3 at $\Delta = 0.1$ gives a mean OLS estimate of -0.47 but a mean NLLS estimate of 0.01 (Table 1, Panel C)
- 15-minute steps, $h = 20$: MSE ($\times 10^8$) 0.038 for log-HAR, 0.020 for RFSV, 0.019 for Power-BSS (Table 4, Panel A); daily steps, $h = 1$: RFSV is best, 0.493 ($\times 10^9$) (Table 5, Panel B): modelling persistence pays intraday and at long horizons

Case Study: Replicate, Limits and an Open Extension

- **Replicate** on data/market/intraday/SPY.US_5m.csv (5-minute bars, 2020–2026, 09:30–16:00 New York time): $\hat{\sigma}_t^2 =$ bipower variation [Barndorff-Nielsen & Shephard \(2004\)](#) of the 5-minute returns in each bin of $\Delta = 30$ minutes, 65 minutes or 1 day, divided by Δ and by the seasonal factor
 - ▶ OLS and NLLS $\hat{\alpha}$, Cauchy $\hat{\beta}^*$ and Gamma-BSS $\hat{\lambda}^*$ against Table 3, Panel A; log-HAR, RFSV and BSS forecasts, MSE, QLIKE and MCS against Table 5
 - ▶ Data limit: bars instead of every trade, so the pre-averaged estimator (Appendix A) is not feasible and the noise is larger; only 13 returns per 65-minute bin
- **Limits stated by the authors**
 - ▶ Realised measures are noisy estimates of integrated variance: the conclusions concern them, not the latent spot volatility (Sections 5.2 and 7)
 - ▶ No rule for the NLLS bandwidth m (Section 4.1); NLLS is too costly for rolling forecasts, which use the OLS $\hat{\alpha}$ (Section 6.1.1)
 - ▶ Forecasts assume a Gaussian X ; a non-Gaussian volatility of volatility is left out of scope (Section 6, footnote 3)
- **Open extension:** why does long memory improve even short-horizon forecasts? The authors leave it to further research (Section 6.1.2)
 - ▶ Rolling $\hat{\alpha}$ (65 minutes, 2005–2014, median -0.36) is less rough in 2008 and 2011 (Figure 3), and $\hat{\alpha}$ rises with log volume, slope 0.0091 (Figure 6): does Bitcoin (BTC-USD.CC_5m.csv), a market that never closes, fit this pattern?

An Open Question: How Much of the Price Path Is Discontinuous?

- Model: $dX_t = \mu_t dt + \sigma_t dW_t + J_t dN_t$; the answer depends on the sampling frequency and on the model
 - ▶ Daily Merton MLE (this chapter): jumps carry 71% of S&P 500 variance; the 5-minute bipower test (Chapter 9): 2.8% for SPY
 - ▶ Christensen, Oomen & Podolskij (2014): at tick frequency most jumps detected at coarser sampling disappear; they are rapid but continuous moves
 - ▶ Christensen, Oomen & Renò (2022): **drift bursts**, short explosive trends in μ_t , look like jumps in coarse data
- Formal question: $H_0: \lambda = 0$ (continuous path with stochastic volatility) against $H_1: \lambda > 0$
 - ▶ Non-standard: $\lambda = 0$ lies on the boundary and (μ_J, σ_J) are unidentified under H_0 , so the LR statistic is not χ^2

The Discovery Loop, Step by Step (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: tabulate estimates by frequency, then check every DOI on [Crossref](#) (the title must match)
 - ▶ Tabulate published estimates of the jump share of equity-index variance by sampling frequency (daily, 5-minute, tick) and method; give the exact number, the table it comes from, and the DOI.
- 2. **Competing hypotheses**, with distinct testable implications
 - ▶ For three processes that give a daily excess kurtosis near 11 (Merton jumps, Heston volatility, drift bursts), derive the implications for returns standardised by ex-ante volatility, for bipower versus realised variance, and for the drift just before extreme moves.
- 3. **Data and code**: replicate the chapter's estimate first
 - ▶ Write Python that reads `data/market/GSPC.INDX.csv` and fits the Merton model by maximum likelihood (Poisson mixture truncated at 12 jumps, several starting points). Target: $\hat{\lambda} = 102$ per year, jump share 71%

The Discovery Loop, Step by Step (2/2)

- 4. **Robustness**, pre-registered
 - ▶ Write a pre-registered plan: standardise returns by EWMA and by GARCH volatility known at $t-1$; split 1990-2007 and 2008-2026; test Heston (no jumps, the null) against Bates (SVJ); obtain LR critical values by parametric bootstrap under the Heston null.
- 5. **Critique**: ask the assistant to attack the inference
 - ▶ Act as a referee: why is the LR test of $\lambda = 0$ non-standard, what is the correct reference distribution, and could a Poisson mixture with frequent small jumps simply be a fat-tailed innovation?
- 6. **Report**: all frequencies and filters, bootstrap critical values, code and the AI log

What the Researcher Must Check

- **Boundary and unidentified parameters:** the LR for $\lambda = 0$ has no χ^2 limit; test: parametric bootstrap under H_0
- **Look-ahead in the filter:** a two-sided (smoothed) volatility absorbs jumps artificially; test: only $\mathcal{F}_{t-1}$ in the standardisation
- **Frequency mismatch:** daily and 5-minute jump shares answer different questions; test: report the share at every frequency on a common sample
- **Multimodal likelihood:** a Poisson mixture has local optima; test: many starting points and a profile likelihood in λ
- **Discretisation:** Euler steps on a coarse grid create spurious large moves; test: exact simulation where it exists
- **Invented numbers:** claims such as “jumps explain 10% of variance” need a table and a DOI; check both

Mini-Case: Jumps or Stochastic Volatility in Disguise?

- Replication: Merton on S&P 500 daily returns, $\hat{\lambda} = 102$ per year, jump share 71% (Section 9)
- A typical assistant proposal: “71% of variance comes from jumps”. Rival hypothesis: the Poisson mixture absorbs time-varying volatility; then standardising by EWMA volatility ($\lambda_{\text{EWMA}} = 0.94$, known at $t - 1$) removes the jumps

	Raw returns	EWMA-standardised
Excess kurtosis	10.9	2.9
LR, Merton vs the Normal distribution	2 401	619
$\hat{\lambda}$ per year (s.e.)	102 (11)	124 (53)
Jump share of variance	71%	51%

- Standardising removes most kurtosis, yet the model still reports a “jump” about every other day: $\hat{\lambda}$ measures fat tails, not discontinuities
- Interpretation: daily data do not identify the jump share; Lee–Mykland finds 0.87 jumps per year and 5-minute data give 2.8%. Code: `Quantlets/Ch_11/ai_discovery_case.py`

Your Turn: A Project Seed

- Question: how much of S&P 500 and Bitcoin variance is discontinuous, at which frequency, and are the detected “jumps” drift bursts?
- Replicate first: Merton jump share 71% (daily S&P 500), Lee–Mykland 0.87 jumps per year, bipower jump share 2.8% (SPY, 5 minutes, Chapter 9)
- Deliverables
 - ▶ A pre-registration (hypotheses, frequencies, filters, tests, bootstrap design) dated before the first run
 - ▶ The jump share at 5, 15, 30 minutes and daily; the drift-burst statistic of Christensen, Oomen & Renò (2022) around April 2025; a bootstrap LR test of Heston (null, no jumps) against Bates
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in AI_USE.md and every error the assistant made in AI_ERRORS.md: github.com/danpele/MFM/tree/main/project

Summary

- Brownian motion is the universal limit of random walks; its quadratic variation t drives Itô calculus
- Itô's lemma gives explicit solutions (GBM, Ornstein–Uhlenbeck, Merton); simulate exactly or with Euler–Maruyama and Milstein
- Identification: infill pins down volatility; drift and mean-reversion speed need long spans, and $\hat{\kappa}$ is biased
- Girsanov changes only the drift; Feynman–Kac turns a Q -expectation into a PDE; the VIX is a Q -object, affine in v_t
- Data: in our fits jumps help with the tails and stochastic volatility with the clustering; the literature favours models with both

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Applied Computational Finance
- ▣  XFG Advanced Methods in Quantitative Finance

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

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